

VAPO: Value-Model-Based RL for Advanced Reasoning

60.4 on AIME 2024 | Surpasses DAPO by 10+ Points | Qwen2.5-32B Base, No SFT

ByteDance Seed · arXiv:2504.05118

Why Value Models? Finer Credit Assignment for Long Chain-of-Thought Reasoning

Value-Model-Free (GRPO, DAPO)

- Compute advantage from final trajectory reward
- Assign same advantage to every token uniformly
- High variance for long, heterogeneous CoT lengths
- No fine-grained step-level error attribution

Value-Model-Based (VAPO)

- Token-level credit assignment via learned value model
- Lower-variance advantage estimates per token
- Generalizes across samples → better sample efficiency
- Critical when subtle step errors cause catastrophic failure

Three Core Challenges in Value-Model-Based Long-CoT RL

Long chain-of-thought reasoning exposes fundamental weaknesses in standard value estimation

1. Value Model Bias

Long trajectories and bootstrapped learning amplify bias in value estimates, especially early in training.

2. Heterogeneous Sequence Lengths

Responses vary from hundreds to thousands of tokens; a single bias-variance tradeoff cannot fit all lengths.

3. Reward Signal Sparsity

Only the final <eos> token receives reward, making credit assignment through long reasoning chains extremely hard.

VAPO Framework: Integrated Solutions for Value-Model-Based Reasoning RL

Each component maps directly to one of the three main challenges.

Challenge 1: Value Model Bias

Value-Pretraining

Pre-trains value model before RL to reduce initial bias.

Decoupled-GAE

Separates GAE computation for more robust value estimation (from VC-PPO).

Challenge 2: Heterogeneous Lengths

Length-Adaptive GAE

Dynamically adjusts λ based on response length: short $\rightarrow$ low λ , long $\rightarrow$ high λ .

Clip-Higher

Asymmetric clipping encourages exploration (from DAPO).

Challenge 3: Reward Sparsity

Token-level Loss

Normalizes loss per token rather than per sequence (from DAPO).

Positive Example LM Loss

Self-imitation learning on correct trajectories.

Group-Sampling

Multiple responses per prompt for better baseline (from GRPO).

Length-Adaptive GAE: Dynamic λ Adjustment for Variable Response Lengths

Novel technique that optimizes the bias-variance tradeoff across heterogeneous sequence lengths

Core Idea

- Standard GAE uses a fixed λ for all sequences.
- Long-CoT responses vary wildly in length (hundreds $\rightarrow$ thousands of tokens).
- Short responses benefit from lower $\lambda \rightarrow$ lower variance (reliable but biased).
- Long responses benefit from higher $\lambda \rightarrow$ lower bias (more accurate credit).
- Length-Adaptive GAE dynamically sets $\lambda = f(\text{length})$ per sample.

Short Response

Lower $\lambda \rightarrow$ lower variance
More stable value estimates
Tolerable bias for short chains

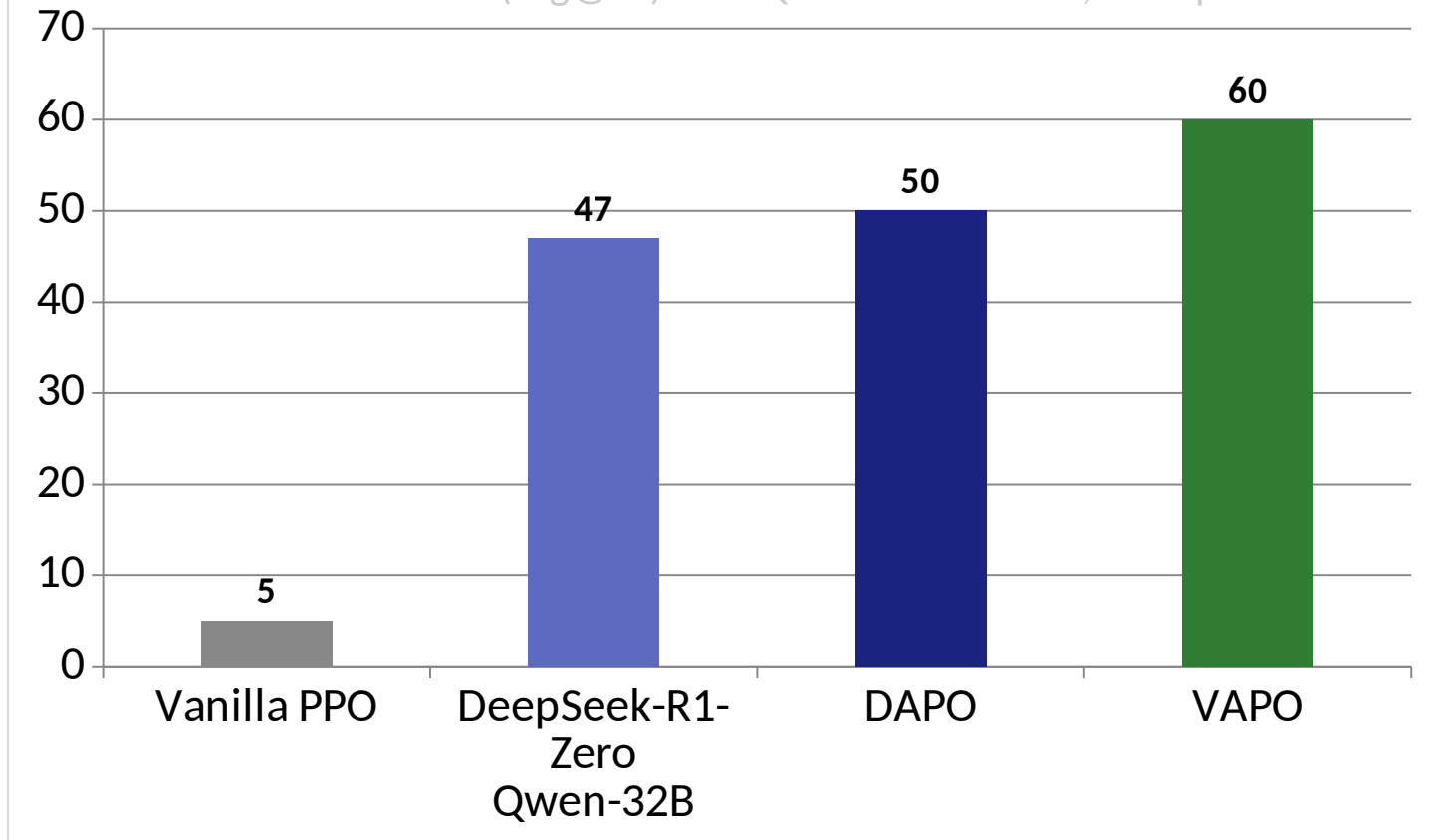
Long Response

Higher $\lambda \rightarrow$ lower bias
More accurate credit assignment
Critical for long reasoning chains

Impact: Removing Length-Adaptive GAE drops AIME score from 60 $\rightarrow$ 45

VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO and DeepSeek-R1-Zero

Main results on AIME 2024 (avg@32) with Qwen2.5-32B base, no supervised fine-tuning



Key Results

- VAPO: 60.4 — new SOTA on AIME 2024
 - DAPO: 50 — previous best value-free method
 - DeepSeek-R1-Zero: 47 — same base model
 - Vanilla PPO: 5 — baseline collapse
-
- VAPO reaches 60% accuracy by step ~4,500
 - DAPO plateaus near 50% at step ~5,500
 - Zero training crashes across multiple runs

→ See training curves on next slide (figure_2.jpg)

VAPO Reaches SOTA in ~5K Steps — Faster and More Stable Than DAPO

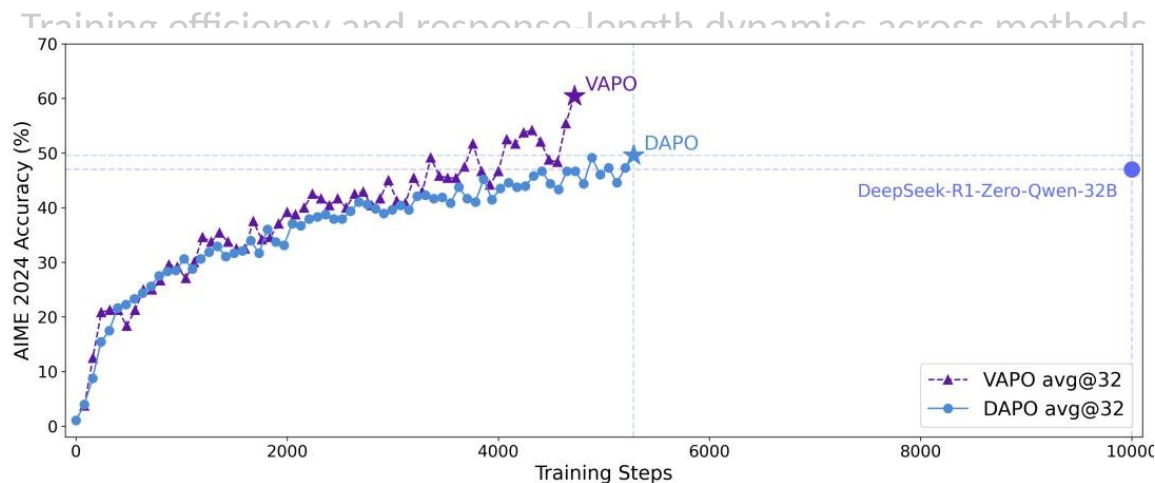


Figure 2: VAPO vs DAPO accuracy over training steps

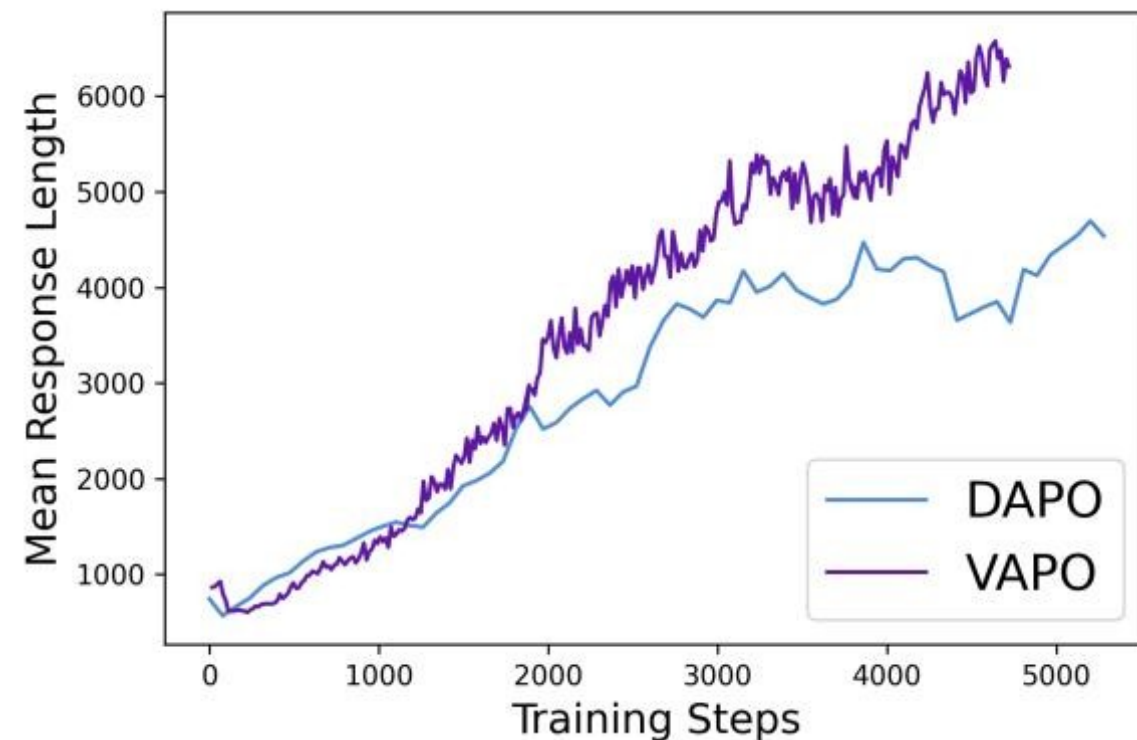


Figure 3: Mean response length over training

Insight: VAPO generates longer reasoning chains (~6,000 tokens) than DAPO (~4,500), suggesting more elaborate exploration without collapse.

Ablation: Every Component Matters — Value-Pretraining Alone Adds 49 Points

Component contribution measured by AIME 2024 score drop when removed from full VAPO (60)

Configuration	AIME 2024 Score	Δ from VAPO	Relative Impact
VAPO (full)	60	—	Baseline
w/o Value-Pretraining	11	-49	Critical — value model collapses
w/o Decoupled-GAE	33	-27	Major — unstable value estimation
w/o Length-Adaptive GAE	45	-15	Strong — heterogeneous length mismatch
w/o Clip-Higher	46	-14	Moderate — reduced exploration
w/o Token-level Loss	53	-7	Minor — sequence-level normalization harm
w/o Positive Example LM Loss	54	-6	Minor — less self-imitation signal

Takeaway: Value-Pretraining is the single most important component. Without it, value-model-based RL barely outperforms Vanilla PPO.

Key Takeaways: Value-Model-Based RL Is Back — With the Right Engineering

Implications for the next generation of reasoning model training

1

1. Value-model-based RL can surpass value-free methods

When bias, heterogeneous lengths, and sparsity are systematically addressed, token-level credit assignment wins.

2

2. Length-Adaptive GAE is critical for long-CoT reasoning

A fixed λ cannot serve sequences ranging from hundreds to thousands of tokens; dynamic adjustment adds 15 points.

3

3. Value-Pretraining is the single most important component

It contributes 49 points (60 $\rightarrow$ 11). Without it, the value model collapses and the framework fails.

4

4. VAPO is highly training-efficient and stable

Reaches SOTA in $\sim 5,000$ steps with zero crashes, versus DAPO's $\sim 5,500$ steps for a lower score.

5

5. Implications for future reasoning models

Engineering around value-model challenges unlocks finer credit assignment, enabling longer and more reliable reasoning chains.